

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	LTVIP2026TMIDS74350
Project Name	Measuring the pulse of prosperity: An Index of economic freedom analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Input & Configuration	• The system shall allow selection of countries or regions • The system shall allow selection of analysis period (single year or multiple years) • The system shall allow selection of economic freedom dimensions • The system shall allow selection of prosperity indicators
FR-2	Data Upload & Integration	• The system shall accept Economic Freedom Index datasets • The system shall accept economic indicator datasets • The system shall support CSV and Excel file formats • The system shall allow multiple dataset uploads
FR-3	Data Validation	• The system shall detect missing values • The system shall detect duplicate entries • The system shall verify country-year consistency • The system shall notify the user of data issues
FR-4	Data Pre-processing	• The system shall clean invalid or inconsistent data • The system shall normalize numerical values • The system shall align datasets by country and year • The system shall store processed data

FR-5	Economic Freedom Analysis	• The system shall calculate overall economic freedom scores
		• The system shall analyze component-wise scores • The system shall rank countries by freedom level • The system shall classify countries as high, medium, or low freedom
FR-6	Prosperity Indicators Analysis	• The system shall analyze GDP per capita trends • The system shall analyze employment and unemployment rates • The system shall analyze poverty and living standards • The system shall compare prosperity across regions
FT-7	Correlation & Comparative Analysis	• The system shall compute correlation between freedom and prosperity • The system shall perform country-to-country comparison • The system shall perform region-wise comparison • The system shall highlight significant relationships
FT-8	Visualization & Reporting	• The system shall generate graphs and charts • The system shall generate summary tables • The system shall export reports in PDF or PPT format • The system shall store generated reports

FR No.	Non-Functional Requirement	Description
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NFR-1	Usability	• The system shall have a user-friendly interface
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Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

		• The system shall require minimal training • The system shall provide clear labels and instructions
NFR-2	Security	• The system shall restrict unauthorized access • The system shall protect stored datasets • The system shall ensure secure data handling
NFR-3	Reliability	• The system shall operate without failure during analysis • The system shall prevent data loss • The system shall recover gracefully from errors
NFR-4	Performance	• The system shall process datasets within acceptable time • The system shall efficiently handle large datasets • The system shall maintain performance during repeated runs
NFR-5	Availability	• The system shall be available when required • The system shall minimize downtime
NFR-6	Scalability	• The system shall support addition of new countries • The system shall support additional years of data • The system shall allow inclusion of new indicators

